

Faculty of Information Technology (FIT) – Ho Chi Minh City University of Science (HCMUS)

CS423 / CSC13003 – Software Testing (AI-augmented · 2026)

1. Student information

- **Student name (printed):** Trần Trí Nhân
- **Student ID:** 23127097
- **Class / Cohort:** Kiểm thử phần mềm - 23KTPM1
- **Assignment ID:** HW02
- **Assignment date:** 10/06/2026
- **AI tool(s) used:** ChatGPT
- **AI tool(s) used:**
 - ☒ Yes
 - ☐ No

2. Audit Table

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning
Tool: ChatGPT Time: 21:21 04/06/2026 Input: README.md(from EShop github repo), S04_Domain Testing.pdf(course slides about domain testing) Prompt: "Read FR-01 specification in README.md, try to understand the requirements(conditions) for each input field. Next, for each field and for each requirement(condition) of that field, identify equivalence classes based on guidelines mentioned in S04_Domain Testing slide (Guidelines for range of values, set of values and 'must be' situation)"	Based on FR-01: Đăng ký tài khoản , the input fields are: 1. Họ Tên (Full Name) 2. Email 3. Mật khẩu (Password) 4. Xác nhận mật khẩu (Confirm Password) Using the Domain Testing guidelines: _ Range of values → 1 valid EC + 2 invalid ECs (below range, above range) _ Set of values → one valid EC for each meaningful set + one invalid EC _ Must be condition → 1 valid EC + 1 invalid EC --- # FR-01 Analysis ## 1. Field: Họ Tên (Full Name) ### Requirement R1: User must provide Full Name This is a "must be" condition . EC ID Equivalence Class Validity ----- ----- ----- FN-EC1 Full Name is provided (non-empty) Valid FN-EC2 Full Name is empty/null Invalid --- ## 2. Field: Email ### Requirement R2: User must provide Email (must be) EC ID Equivalence Class Validity ----- ----- ----- EM-EC1 Email provided Valid EM-EC2 Email empty/null Invalid --- ### Requirement R3: Email must have valid format (user@domain.com)	VALID	Identification of equivalence classes from the AI correctly follow the guidelines from course slides(S04_Domain Testing.pdf), while not missing any key requirements from the functional specification(README of EShop).

(1) Prompt + Tool

(2) AI Output

(3) Verdict

(4) Reasoning

This is a **must be format condition**.

EC ID	Equivalence Class	Validity
-----	-----	-----
EM-EC3	Valid email format	Valid
EM-EC4	Invalid email format	Invalid

Examples of EM-EC4:

_ userdomain.com
_ user@
_ @gmail.com
_ abc

Requirement R4: Email must be unique

This is a **set of values condition** because the system treats existing and non-existing emails differently.

EC ID	Equivalence Class	Validity
-----	-----	-----
EM-EC5	Email does not exist in system	Valid
EM-EC6	Email already exists in system	Invalid

3. Field: Password

Requirement R5: User must provide Password

(must be)

EC ID	Equivalence Class	Validity
-----	-----	-----
PW-EC1	Password provided	Valid
PW-EC2	Password empty/null	Invalid

Requirement R6: Password length ≥ 8 characters

This is a **range condition**.

EC ID	Equivalence Class	Validity
-----	-----	-----
PW-EC3	Length ≥ 8	Valid
PW-EC4	Length < 8	Invalid

Since only a lower bound exists, there is effectively one invalid side.

Requirement R7: Password contains at least one uppercase letter

(must be)

EC ID	Equivalence Class	Validity
-----	-----	-----
PW-EC5	Contains ≥ 1 uppercase letter	Valid
PW-EC6	Contains no uppercase letter	Invalid

Requirement R8: Password contains at least one lowercase letter

(1) Prompt + Tool

(2) AI Output

(3) Verdict

(4) Reasoning

(must be)

EC ID	Equivalence Class	Validity
-----	-----	-----
PW-EC7	Contains ≥1 lowercase letter	Valid
PW-EC8	Contains no lowercase letter	Invalid

Requirement R9: Password contains at least one digit

(must be)

EC ID	Equivalence Class	Validity
-----	-----	-----
PW-EC9	Contains ≥1 digit	Valid
PW-EC10	Contains no digit	Invalid

Requirement R10: Password contains at least one special character

Allowed special characters:

@ \$! % _ ? &

This is a **set of values + must be condition**.

EC ID	Equivalence Class	Validity
-----	-----	-----
PW-EC11	Contains at least one allowed special character	Valid
PW-EC12	Contains no special character	Invalid
PW-EC13	Contains special character not in allowed set only	Invalid

Examples:

* Abc123@ → PW-EC11

Abc12345 → PW-EC12

Abc123# → PW-EC13

4. Field: Confirm Password

Requirement R11: Confirm Password must be provided

(must be)

EC ID	Equivalence Class	Validity
-----	-----	-----
CP-EC1	Confirm Password provided	Valid
CP-EC2	Confirm Password empty/null	Invalid

Requirement R12: Confirm Password must match Password

This is a **set-of-values relationship condition**.

EC ID	Equivalence Class	Validity
-----	-----	-----
CP-EC3	Confirm Password = Password	Valid
CP-EC4	Confirm Password ≠ Password	Invalid

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning																																																				
	# Consolidated Equivalence Classes <table><tr><th>Field</th><th>Requirement</th><th>Valid ECs</th><th>Invalid ECs</th></tr><tr><td>Full Name</td><td>Required</td><td>FN-EC1</td><td>FN-EC2</td></tr><tr><td>Email</td><td>Required</td><td>EM-EC1</td><td>EM-EC2</td></tr><tr><td>Email</td><td>Format</td><td>EM-EC3</td><td>EM-EC4</td></tr><tr><td>Email</td><td>Unique</td><td>EM-EC5</td><td>EM-EC6</td></tr><tr><td>Password</td><td>Required</td><td>PW-EC1</td><td>PW-EC2</td></tr><tr><td>Password</td><td>Length ≥ 8</td><td>PW-EC3</td><td>PW-EC4</td></tr><tr><td>Password</td><td>Uppercase</td><td>PW-EC5</td><td>PW-EC6</td></tr><tr><td>Password</td><td>Lowercase</td><td>PW-EC7</td><td>PW-EC8</td></tr><tr><td>Password</td><td>Digit</td><td>PW-EC9</td><td>PW-EC10</td></tr><tr><td>Password</td><td>Special char</td><td>PW-EC11</td><td>PW-EC12, PW-EC13</td></tr><tr><td>Confirm Password</td><td>Required</td><td>CP-EC1</td><td>CP-EC2</td></tr><tr><td>Confirm Password</td><td>Match Password</td><td>CP-EC3</td><td>CP-EC4</td></tr></table> These equivalence classes follow the Domain Testing partitioning rules presented in the S04 slides: identify each input condition separately and create valid/invalid partitions according to range, set-of-values, and must-be constraints.	Field	Requirement	Valid ECs	Invalid ECs	Full Name	Required	FN-EC1	FN-EC2	Email	Required	EM-EC1	EM-EC2	Email	Format	EM-EC3	EM-EC4	Email	Unique	EM-EC5	EM-EC6	Password	Required	PW-EC1	PW-EC2	Password	Length ≥ 8	PW-EC3	PW-EC4	Password	Uppercase	PW-EC5	PW-EC6	Password	Lowercase	PW-EC7	PW-EC8	Password	Digit	PW-EC9	PW-EC10	Password	Special char	PW-EC11	PW-EC12, PW-EC13	Confirm Password	Required	CP-EC1	CP-EC2	Confirm Password	Match Password	CP-EC3	CP-EC4		
Field	Requirement	Valid ECs	Invalid ECs																																																				
Full Name	Required	FN-EC1	FN-EC2																																																				
Email	Required	EM-EC1	EM-EC2																																																				
Email	Format	EM-EC3	EM-EC4																																																				
Email	Unique	EM-EC5	EM-EC6																																																				
Password	Required	PW-EC1	PW-EC2																																																				
Password	Length ≥ 8	PW-EC3	PW-EC4																																																				
Password	Uppercase	PW-EC5	PW-EC6																																																				
Password	Lowercase	PW-EC7	PW-EC8																																																				
Password	Digit	PW-EC9	PW-EC10																																																				
Password	Special char	PW-EC11	PW-EC12, PW-EC13																																																				
Confirm Password	Required	CP-EC1	CP-EC2																																																				
Confirm Password	Match Password	CP-EC3	CP-EC4																																																				
Tool: ChatGPT Time: 21:31 04/06/2026 Prompt: "Now design the test cases to cover these classes. You should follow the guideline mentioned in S04 Domain testing slide: Choose at least one test case from each equivalence class. For valid classes, choose test cases to cover as many equivalence classes as possible, until all valid classes have been covered. For invalid classes, choose test cases so that each covers one and only one invalid class, until all classes are covered"	Following the slide guideline: > For valid classes, combine as many valid ECs as possible into a minimum number of test cases. > > For invalid classes, each test case should cover one and only one invalid EC while all other conditions remain valid. --- # Step 1. Create a Baseline Valid Test Case Choose values satisfying all valid ECs simultaneously: <table><tr><th>Field</th><th>Value</th></tr><tr><td>Full Name</td><td>Nguyen Van A</td></tr><tr><td>Email</td><td>newuser@gmail.com</td></tr><tr><td>Password</td><td>Abc1234@</td></tr><tr><td>Confirm Password</td><td>Abc1234@</td></tr></table> Covered ECs: _ FN-EC1 _ EM-EC1 _ EM-EC3 _ EM-EC5 _ PW-EC1 _ PW-EC3 _ PW-EC5 _ PW-EC7 _ PW-EC9 _ PW-EC11 _ CP-EC1 _ CP-EC3 --- # Test Cases <table><tr><th>TC ID</th><th>Full Name</th><th>Email</th><th>Password</th><th>Confirm Password</th><th>EC Covered</th><th>Expected Result</th></tr><tr><td>TC01</td><td>Nguyen Van A</td><td>newuser@gmail.com</td><td>Abc1234@</td><td>Abc1234@</td><td>All valid ECs</td><td>Registration successful, redirect to Login</td></tr></table>	Field	Value	Full Name	Nguyen Van A	Email	newuser@gmail.com	Password	Abc1234@	Confirm Password	Abc1234@	TC ID	Full Name	Email	Password	Confirm Password	EC Covered	Expected Result	TC01	Nguyen Van A	newuser@gmail.com	Abc1234@	Abc1234@	All valid ECs	Registration successful, redirect to Login	INCOMPLETE	The AI correctly follow the the guidelines(S04_Domain Testing.pdf) to create a minimal set of test case for valid classes and invalid classes from the previous stage. However, the AI miss the objectives and the steps of the test cases.																												
Field	Value																																																						
Full Name	Nguyen Van A																																																						
Email	newuser@gmail.com																																																						
Password	Abc1234@																																																						
Confirm Password	Abc1234@																																																						
TC ID	Full Name	Email	Password	Confirm Password	EC Covered	Expected Result																																																	
TC01	Nguyen Van A	newuser@gmail.com	Abc1234@	Abc1234@	All valid ECs	Registration successful, redirect to Login																																																	

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning
	--- # Invalid Equivalence Classes ## FN-EC2 (Full Name empty) TC ID Full Name Email Password Confirm Password EC Covered Expected ---- - - - - - -- - - TC02 "" newuser@gmail.com Abc1234@ Abc1234@ FN-EC2 Error: Full Name required --- ## EM-EC2 (Email empty) TC ID Full Name Email Password Confirm Password EC Covered Expected ---- - - - - - TC03 Nguyen Van A "" Abc1234@ Abc1234@ EM-EC2 Error: Email required --- ## EM-EC4 (Invalid email format) TC ID Full Name Email Password Confirm Password EC Covered Expected ---- - - - - - ----- TC04 Nguyen Van A abcgmail.com Abc1234@ Abc1234@ EM-EC4 Error: Invalid email format --- ## EM-EC6 (Email already exists) Assume: test@eshop.com already exists. TC ID Full Name Email Password Confirm Password EC Covered Expected ---- - - - - - ----- TC05 Nguyen Van A test@eshop.com Abc1234@ Abc1234@ EM-EC6 Error: Email already exists --- ## PW-EC2 (Password empty) TC ID Full Name Email Password Confirm Password EC Covered Expected ---- - - - - - ----- TC06 Nguyen Van A newuser@gmail.com "" "" PW-EC2 Error: Password required --- ## PW-EC4 (Length < 8) Everything else remains valid.		

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning
	TC ID Full Name Email Password Confirm Password EC Covered Expected ---- ----- ----- ----- ----- ---- ----- ----- TC07 Nguyen Van A newuser@gmail.com Abc1@ Abc1@ PW-EC4 Error: Password too short Length = 5 --- ## PW-EC6 (No uppercase) TC ID Full Name Email Password Confirm Password EC Covered Expected ---- ----- ----- ----- ----- ---- ----- ----- TC08 Nguyen Van A newuser@gmail.com abc1234@ abc1234@ PW-EC6 Error: Missing uppercase letter --- ## PW-EC8 (No lowercase) TC ID Full Name Email Password Confirm Password EC Covered Expected ---- ----- ----- ----- ----- ---- ----- ----- TC09 Nguyen Van A newuser@gmail.com ABC1234@ ABC1234@ PW-EC8 Error: Missing lowercase letter --- ## PW-EC10 (No digit) TC ID Full Name Email Password Confirm Password EC Covered Expected ---- ----- ----- ----- ----- ---- ----- ----- TC10 Nguyen Van A newuser@gmail.com Abcdefg@ Abcdefg@ PW-EC10 Error: Missing digit --- ## PW-EC12 (No special character) TC ID Full Name Email Password Confirm Password EC Covered Expected ---- ----- ----- ----- ----- ---- ----- ----- TC11 Nguyen Van A newuser@gmail.com Abc12345 Abc12345 PW-EC12 Error: Missing special character --- ## PW-EC13 (Special character not in allowed set) Assuming the specification restricts special characters to: @ \$! % * ? & TC ID Full Name Email Password Confirm Password EC Covered Expected ---- ----- ----- ----- ----- ---- ----- ----- TC12 Nguyen Van A newuser@gmail.com Abc1234# Abc1234# PW-EC13 Error: Invalid special character		

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning																																																																																																																													
	--- ## CP-EC2 (Confirm Password empty) Password remains valid. <table><tr><td>TC ID</td><td>Full Name</td><td>Email</td><td>Password</td><td>Confirm</td></tr><tr><td>Password</td><td>EC Covered</td><td>Expected</td><td></td><td></td></tr><tr><td>----</td><td>-----</td><td>-----</td><td>-----</td><td>-----</td></tr><tr><td>----</td><td>-----</td><td>-----</td><td></td><td></td></tr><tr><td>TC13</td><td>Nguyen Van A</td><td>newuser@gmail.com</td><td>Abc1234@</td><td>""</td></tr><tr><td></td><td></td><td></td><td></td><td>CP-EC2</td></tr><tr><td></td><td></td><td></td><td>Error: Confirm Password required</td><td></td></tr></table> --- ## CP-EC4 (Password mismatch) <table><tr><td>TC ID</td><td>Full Name</td><td>Email</td><td>Password</td><td>Confirm</td></tr><tr><td>Password</td><td>EC Covered</td><td>Expected</td><td></td><td></td></tr><tr><td>----</td><td>-----</td><td>-----</td><td>-----</td><td>-----</td></tr><tr><td>----</td><td>-----</td><td>-----</td><td></td><td></td></tr><tr><td>TC14</td><td>Nguyen Van A</td><td>newuser@gmail.com</td><td>Abc1234@</td><td>Abc12345@</td></tr><tr><td></td><td></td><td></td><td></td><td>CP-EC4</td></tr><tr><td></td><td></td><td></td><td>Error: Passwords do not match</td><td></td></tr></table> --- # Coverage Matrix <table><tr><td>EC</td><td>Covered By</td></tr><tr><td>-----</td><td>-----</td></tr><tr><td>FN-EC1</td><td>TC01</td></tr><tr><td>FN-EC2</td><td>TC02</td></tr><tr><td>EM-EC1</td><td>TC01</td></tr><tr><td>EM-EC2</td><td>TC03</td></tr><tr><td>EM-EC3</td><td>TC01</td></tr><tr><td>EM-EC4</td><td>TC04</td></tr><tr><td>EM-EC5</td><td>TC01</td></tr><tr><td>EM-EC6</td><td>TC05</td></tr><tr><td>PW-EC1</td><td>TC01</td></tr><tr><td>PW-EC2</td><td>TC06</td></tr><tr><td>PW-EC3</td><td>TC01</td></tr><tr><td>PW-EC4</td><td>TC07</td></tr><tr><td>PW-EC5</td><td>TC01</td></tr><tr><td>PW-EC6</td><td>TC08</td></tr><tr><td>PW-EC7</td><td>TC01</td></tr><tr><td>PW-EC8</td><td>TC09</td></tr><tr><td>PW-EC9</td><td>TC01</td></tr><tr><td>PW-EC10</td><td>TC10</td></tr><tr><td>PW-EC11</td><td>TC01</td></tr><tr><td>PW-EC12</td><td>TC11</td></tr><tr><td>PW-EC13</td><td>TC12</td></tr><tr><td>CP-EC1</td><td>TC01</td></tr><tr><td>CP-EC2</td><td>TC13</td></tr><tr><td>CP-EC3</td><td>TC01</td></tr><tr><td>CP-EC4</td><td>TC14</td></tr></table> This is the minimum equivalence-partition test suite that follows the S04 rule: 1 valid test case covering all valid ECs + 1 test case for each invalid equivalence class, with exactly one invalid EC per test case.	TC ID	Full Name	Email	Password	Confirm	Password	EC Covered	Expected			----	-----	-----	-----	-----	----	-----	-----			TC13	Nguyen Van A	newuser@gmail.com	Abc1234@	""					CP-EC2				Error: Confirm Password required		TC ID	Full Name	Email	Password	Confirm	Password	EC Covered	Expected			----	-----	-----	-----	-----	----	-----	-----			TC14	Nguyen Van A	newuser@gmail.com	Abc1234@	Abc12345@					CP-EC4				Error: Passwords do not match		EC	Covered By	-----	-----	FN-EC1	TC01	FN-EC2	TC02	EM-EC1	TC01	EM-EC2	TC03	EM-EC3	TC01	EM-EC4	TC04	EM-EC5	TC01	EM-EC6	TC05	PW-EC1	TC01	PW-EC2	TC06	PW-EC3	TC01	PW-EC4	TC07	PW-EC5	TC01	PW-EC6	TC08	PW-EC7	TC01	PW-EC8	TC09	PW-EC9	TC01	PW-EC10	TC10	PW-EC11	TC01	PW-EC12	TC11	PW-EC13	TC12	CP-EC1	TC01	CP-EC2	TC13	CP-EC3	TC01	CP-EC4	TC14			
TC ID	Full Name	Email	Password	Confirm																																																																																																																												
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EM-EC5	TC01																																																																																																																															
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(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning									
Tool: ChatGPT Time: 21:50 04/06/2026 Propmt: "Next, I want you to use Boundary Value Analysis, follow the guidelines in S04 domain testing(if mentioned), add test cases upon the existed test cases in the previous step."	Yes. After Equivalence Partitioning, S04 introduces Boundary Value Analysis (BVA) because errors are most likely to occur at the edges of ranges. The slide recommends testing around each boundary using: <pre>_ LB-1 _ LB _ LB+1 _ UB-1 _ UB _ UB+1</pre> for every range-based equivalence class. --- # Step 1: Identify Boundary Conditions in FR-01 Among all FR-01 requirements, only one requirement contains a numeric range: ### Password Length Requirement: > Password must contain at least 8 characters. This creates: <pre>_ Invalid EC: Length < 8 _ Valid EC: Length ≥ 8</pre> Boundary: <pre>_ LB = 8</pre> Since no upper bound is specified, only the lower boundary exists. Following S04 BVA: <table><tr><td> Position Length </td></tr><tr><td> ----- ----- </td></tr><tr><td> LB-1 7 </td></tr><tr><td> LB 8 </td></tr><tr><td> LB+1 9 </td></tr></table> --- # Step 2: Existing Coverage Check Current suite already contains: <table><tr><td> TC Password Length </td></tr><tr><td> ---- ----- ----- </td></tr><tr><td> TC01 Abc1234@ 8 </td></tr><tr><td> TC07 Abc1@ 5 </td></tr></table> So: <pre>_ LB already covered (8) _ Below-boundary class covered (5)</pre> But BVA requires testing values immediately adjacent to the boundary: <pre>_ 7 _ 8 _ 9</pre> Therefore additional test cases are needed.	Position Length	----- -----	LB-1 7	LB 8	LB+1 9	TC Password Length	---- ----- -----	TC01 Abc1234@ 8	TC07 Abc1@ 5	INCOMPLETE	The AI correctly follow the guidelines(S04_Domain Testing.pdf) to apply Boundary Value Analysis to create test cases. However, the AI misses the test objective and test step of the test cases.
Position Length												
----- -----												
LB-1 7												
LB 8												
LB+1 9												
TC Password Length												
---- ----- -----												
TC01 Abc1234@ 8												
TC07 Abc1@ 5												

(1) Prompt + Tool

(2) AI Output

(3) Verdict

(4) Reasoning

Additional Boundary Test Cases

Use all other fields valid:

_ Full Name = Nguyen Van A

_ Email = unique valid email

_ Confirm Password = Password

BVA-TC01 — LB-1

Length = 7

Password:

Abc123@

Count:

A b c 1 2 3 @ = 7 chars

Field	Value
Full Name	Nguyen Van A
Email	bva1@gmail.com
Password	Abc123@
Confirm Password	Abc123@

Partitions Covered:

_ PW-EC4 (Length < 8)

Expected:

_ Registration rejected

_ Password length validation error

BVA-TC02 — LB

Length = 8

Password:

Abc1234@

Count = 8

Field	Value
Full Name	Nguyen Van A
Email	bva2@gmail.com
Password	Abc1234@
Confirm Password	Abc1234@

Partitions Covered:

_ PW-EC3 (Length ≥ 8)

Expected:

_ Registration successful

(1) Prompt + Tool

(2) AI Output

(3) Verdict

(4) Reasoning

BVA-TC03 — LB+1

Length = 9

Password:

Abc12345@

Count = 9

Field	Value
Full Name	Nguyen Van A
Email	bva3@gmail.com
Password	Abc12345@
Confirm Password	Abc12345@

Partitions Covered:

_ PW-EC3 (Length ≥ 8)

Expected:

_ Registration successful

Optional Boundary Analysis for "At Least One"

Although not a numeric range in the strict sense, password complexity requirements introduce implicit count boundaries:

Uppercase Letters

Requirement:

≥ 1 uppercase letter

Boundary:

Value	Meaning
0	Invalid
1	Valid
2+	Valid

Additional tests:

TC	Password	Uppercase Count	Expected
BVA-TC04	abc1234@	0	Reject
BVA-TC05	Abc1234@	1	Accept
BVA-TC06	ABc1234@	2	Accept

Lowercase Letters

Boundary:

0 / 1 / 2+

TC	Password	Lowercase Count	Expected
----	----------	-----------------	----------

10 / 31

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning
	<pre> BVA-TC07 ABC1234@ 0 Reject BVA-TC08 AbC1234@ 1 Accept BVA-TC09 Abc1234@ 2+ Accept --- ## Digits Boundary: 0 / 1 / 2+ TC Password Digit Count Expected ----- ----- ----- ----- BVA-TC10 Abcdefg@ 0 Reject BVA-TC11 Abcdef1@ 1 Accept BVA-TC12 Abc1234@ 2+ Accept --- ### Special Characters Boundary: 0 / 1 / 2+ TC Password Special Count Expected ----- ----- ----- ----- BVA-TC13 Abc12345 0 Reject BVA-TC14 Abc1234@ 1 Accept BVA-TC15 Abc123@! 2 Accept --- # Final BVA Suite to Add Strictly following S04's Boundary Value Analysis (numeric boundary testing), the mandatory additional test cases are: TC ID Password Length Boundary Position Expected ----- ----- ----- ----- BVA-TC01 7 LB-1 Reject BVA-TC02 8 LB Accept BVA-TC03 9 LB+1 Accept The tests BVA-TC04–BVA-TC15 are useful "count-boundary" extensions, but they are not strictly required by the S04 BVA rule because the slide applies BVA primarily to explicit numeric ranges.</pre>		
Tool: ChatGPT Time: 08:32 05/06/2026 Prompt: "Now I want you to repeat step 1 for FR-07, that is understanding the requirements and identifying ECs"	<pre>For FR-07: Giỏ hàng (Shopping Cart), we first identify the inputs/conditions from the specification, then derive equivalence classes using the S04 guidelines: _ Range of values → 1 valid EC + invalid ECs below/above range. _ Set of values → valid EC for each meaningful set + invalid EC. _ Must be → 1 valid EC + 1 invalid EC. --- # FR-07 Requirements FR-07 states: 1. Cart displays product list. 2. Quantity can be adjusted via +/- buttons. 3. Adding the same product increases quantity instead of creating a new row. 4. Delete product requires confirmation dialog. 5. Continue Shopping button returns to home page. 6. Total amount label must be exactly "Tổng cộng".</pre>	INCOMPLETE	The AI correctly follow the guidelines(S04_Domain Testing.pdf) and the specification(README. of EShop) to identify equivalence classes. However, the AI still us total label and continu shopping as requirements to identi equivalence classes, ev though the AI itself pointed out that those requirements were UI checks rather than inpi domain conditions.

(1) Prompt + Tool

(2) AI Output

(3) Verdict

(4) Reasoning

7. Empty cart must display illustration and message.

Input Variables

The requirements involving input/domain behavior are:

- 1. Product being added to cart
- 2. Quantity
- 3. Delete confirmation choice
- 4. Cart state (empty/non-empty)

The other requirements ("Continue Shopping", label text) are UI checks rather than input-domain conditions.

1. Product Addition

Requirement R1

Adding the same product again increases quantity rather than creating a new row.

This is a **set-of-values condition** because the system handles new and existing products differently.

ECs

EC ID	Equivalence Class	Validity
PROD-EC1	Product not currently in cart	Valid
PROD-EC2	Product already exists in cart	Valid

Both are valid because both scenarios are allowed, but they produce different expected behavior.

2. Quantity

Requirement R2

Quantity is editable using +/- controls.

Requirement R3

Quantity must remain positive.

Although FR-07 does not explicitly restate it, FR-06 defines quantity as:

> positive integer, minimum = 1.

Since the cart modifies that quantity, the same constraint applies.

This is a combination of:

Condition A: Must be an integer

EC ID	Equivalence Class	Validity
QTY-EC1	Quantity is integer	Valid
QTY-EC2	Quantity is non-integer	Invalid

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning																																																
	### Condition B: Range (minimum 1) <table><tr><td>EC ID</td><td>Equivalence Class</td><td>Validity</td></tr><tr><td>-----</td><td>-----</td><td>-----</td></tr><tr><td>QTY-EC3</td><td>Quantity ≥ 1</td><td>Valid</td></tr><tr><td>QTY-EC4</td><td>Quantity < 1</td><td>Invalid</td></tr></table> --- # 3. Delete Confirmation Dialog ## Requirement R4 Deleting a product requires confirmation before execution. This is a set-of-values condition. ### ECs <table><tr><td>EC ID</td><td>Equivalence Class</td><td>Validity</td></tr><tr><td>-----</td><td>-----</td><td>-----</td></tr><tr><td>DEL-EC1</td><td>User confirms deletion</td><td>Valid</td></tr><tr><td>DEL-EC2</td><td>User cancels deletion</td><td>Valid</td></tr></table> Both are valid user actions with different outcomes. --- # 4. Cart State ## Requirement R5 Empty cart must display illustration and message. This is a set-of-values condition. ### ECs <table><tr><td>EC ID</td><td>Equivalence Class</td><td>Validity</td></tr><tr><td>-----</td><td>-----</td><td>-----</td></tr><tr><td>CART-EC1</td><td>Cart contains at least one item</td><td>Valid</td></tr><tr><td>CART-EC2</td><td>Cart is empty</td><td>Valid</td></tr></table> Both are legitimate states. --- # 5. Total Label ## Requirement R6 Total label must be exactly "Tổng cộng". This is a must-be condition. ### ECs <table><tr><td>EC ID</td><td>Equivalence Class</td><td>Validity</td></tr><tr><td>-----</td><td>-----</td><td>-----</td></tr><tr><td>LABEL-EC1</td><td>Label = "Tổng cộng"</td><td>Valid</td></tr><tr><td>LABEL-EC2</td><td>Label ≠ "Tổng cộng"</td><td>Invalid</td></tr></table> Examples: _ "Tổng tạm tính" _ "Total" _ "Grand Total"	EC ID	Equivalence Class	Validity	-----	-----	-----	QTY-EC3	Quantity ≥ 1	Valid	QTY-EC4	Quantity < 1	Invalid	EC ID	Equivalence Class	Validity	-----	-----	-----	DEL-EC1	User confirms deletion	Valid	DEL-EC2	User cancels deletion	Valid	EC ID	Equivalence Class	Validity	-----	-----	-----	CART-EC1	Cart contains at least one item	Valid	CART-EC2	Cart is empty	Valid	EC ID	Equivalence Class	Validity	-----	-----	-----	LABEL-EC1	Label = "Tổng cộng"	Valid	LABEL-EC2	Label ≠ "Tổng cộng"	Invalid		
EC ID	Equivalence Class	Validity																																																	
-----	-----	-----																																																	
QTY-EC3	Quantity ≥ 1	Valid																																																	
QTY-EC4	Quantity < 1	Invalid																																																	
EC ID	Equivalence Class	Validity																																																	
-----	-----	-----																																																	
DEL-EC1	User confirms deletion	Valid																																																	
DEL-EC2	User cancels deletion	Valid																																																	
EC ID	Equivalence Class	Validity																																																	
-----	-----	-----																																																	
CART-EC1	Cart contains at least one item	Valid																																																	
CART-EC2	Cart is empty	Valid																																																	
EC ID	Equivalence Class	Validity																																																	
-----	-----	-----																																																	
LABEL-EC1	Label = "Tổng cộng"	Valid																																																	
LABEL-EC2	Label ≠ "Tổng cộng"	Invalid																																																	

(1) Prompt + Tool

(2) AI Output

(3) Verdict

(4) Reasoning

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# 6. Continue Shopping Button

## Requirement R7

Continue Shopping returns user to Home page.

This is essentially a navigation requirement.

### ECs

| EC ID | Equivalence Class          | Validity |
| -----|-----|-----|
| NAV-EC1 | Redirects to Home page      | Valid   |
| NAV-EC2 | Redirects elsewhere / no redirect | Invalid |

---

# Consolidated EC List

| Field / Condition | EC      | Description          | Validity |
| -----|-----|-----|-----|
| Product Addition | PROD-EC1 | Product not in cart   | Valid   |
| Product Addition | PROD-EC2 | Product already in cart | Valid   |
| Quantity Type    | QTY-EC1  | Integer quantity     | Valid   |
| Quantity Type    | QTY-EC2  | Non-integer quantity | Invalid |
| Quantity Range   | QTY-EC3  | Quantity ≥ 1         | Valid   |
| Quantity Range   | QTY-EC4  | Quantity < 1         | Invalid |
| Delete Action    | DEL-EC1  | Confirm deletion     | Valid   |
| Delete Action    | DEL-EC2  | Cancel deletion      | Valid   |
| Cart State       | CART-EC1 | Non-empty cart       | Valid   |
| Cart State       | CART-EC2 | Empty cart           | Valid   |
| Total Label      | LABEL-EC1 | "Tổng cộng"          | Valid   |
| Total Label      | LABEL-EC2 | Any other label      | Invalid |
| Continue Shopping | NAV-EC1  | Goes to Home page    | Valid   |
| Continue Shopping | NAV-EC2  | Does not go to Home page | Invalid |

### Note

Unlike FR-01, FR-07 contains many state/UI behavior requirements rather than pure input validation. Therefore several equivalence classes (PROD-EC1/2, DEL-EC1/2, CART-EC1/2) are all valid classes representing different system states that must be tested because the expected behavior differs, exactly matching the S04 rule:

> Split an equivalence class if there is reason to believe elements are handled differently by the program.
```

Tool: ChatGPT Time: 20:31 07/06/2026 Prompt: "Discard equivalence classes of total label and continue shopping since they are not input-domain conditions. Now, apply step 2, design test cases, for FR-07"	Agreed. After removing LABEL-EC1/2 and NAV-EC1/2 , the remaining ECs are: EC Description ----- ----- PROD-EC1 Product not currently in cart PROD-EC2 Product already exists in cart QTY-EC1 Quantity is integer QTY-EC2 Quantity is non-integer QTY-EC3 Quantity ≥ 1 QTY-EC4 Quantity < 1 DEL-EC1 User confirms deletion DEL-EC2 User cancels deletion CART-EC1 Cart contains item(s) CART-EC2 Cart is empty Following the S04 rule: _ Cover all valid ECs with as few test cases as possible.	INCOMPLETE	The AI correctly follow the guidelines(S04_Domain Testing.pdf) to design test cases, but it misse the test objective of th test cases.
---	---	------------	--

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning																							
	_ For invalid ECs, each test case should cover exactly one invalid EC. --- # Step 1: Valid Test Cases Unlike FR-01, FR-07 has multiple valid behavioral states that cannot all be covered in a single test because some states are mutually exclusive (e.g., empty cart vs non-empty cart). --- ## TC01 – Add a New Product to Empty Cart ### Purpose Cover: _ PROD-EC1 _ QTY-EC1 _ QTY-EC3 _ CART-EC2 → CART-EC1 transition ### Preconditions Cart is empty. ### Steps 1. Open Product A. 2. Set quantity = 1. 3. Click "Add to Cart". ### Expected Result _ Product A appears in cart. _ Quantity = 1. _ One cart row created. _ Cart becomes non-empty. ### EC Covered <table><tr><td> EC</td><td> </td></tr><tr><td> -----</td><td> </td></tr><tr><td> PROD-EC1</td><td> </td></tr><tr><td> QTY-EC1</td><td> </td></tr><tr><td> QTY-EC3</td><td> </td></tr><tr><td> CART-EC2</td><td> </td></tr><tr><td> CART-EC1</td><td> </td></tr></table> --- ## TC02 – Add Same Product Again ### Purpose Cover: _ PROD-EC2 ### Preconditions Cart contains: <table><tr><td> Product</td><td> Qty</td><td> </td></tr><tr><td> -----</td><td> ---</td><td> </td></tr><tr><td> Product A</td><td> 1</td><td> </td></tr></table>	EC		-----		PROD-EC1		QTY-EC1		QTY-EC3		CART-EC2		CART-EC1		Product	Qty		-----	---		Product A	1			
EC																										

PROD-EC1																										
QTY-EC1																										
QTY-EC3																										
CART-EC2																										
CART-EC1																										
Product	Qty																									
-----	---																									
Product A	1																									

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning
	### Steps 1. Open Product A again. 2. Add quantity = 1. ### Expected Result _ No new row created. _ Existing row quantity becomes 2. ### EC Covered EC ----- PROD-EC2 --- ## TC03 – Cancel Deletion ### Purpose Cover: _ DEL-EC2 ### Preconditions Cart contains Product A. ### Steps 1. Click Delete Product A. 2. Confirmation dialog appears. 3. Select Cancel. ### Expected Result _ Product remains in cart. _ Quantity unchanged. ### EC Covered EC ----- DEL-EC2 --- ## TC04 – Confirm Deletion ### Purpose Cover: _ DEL-EC1 ### Preconditions Cart contains Product A. ### Steps 1. Click Delete Product A. 2. Confirmation dialog appears. 3. Select Confirm.		

(1) Prompt + Tool

(2) AI Output

(3) Verdict

(4) Reasoning

```
### Expected Result

_ Product removed from cart.

### EC Covered

| EC   |
| ----- |
| DEL-EC1 |

---

# Invalid Test Cases

Only the quantity input generates invalid equivalence classes.

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## TC05 – Quantity is Non-Integer

### Purpose

Cover:

_ QTY-EC2 only

### Preconditions

Product page opened.

### Steps

Enter:

1.5

as quantity.

### Expected Result

System rejects input.

### EC Covered

| EC   |
| ----- |
| QTY-EC2 |

---

## TC06 – Quantity Less Than 1

### Purpose

Cover:

_ QTY-EC4 only

### Preconditions

Product page opened.

### Steps

Enter:
```

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning
	<pre>0 as quantity. ### Expected Result System rejects input. ### EC Covered EC ----- QTY-EC4 --- # Coverage Matrix EC Covered By ----- ----- PROD-EC1 TC01 PROD-EC2 TC02 QTY-EC1 TC01 QTY-EC2 TC05 QTY-EC3 TC01 QTY-EC4 TC06 DEL-EC1 TC04 DEL-EC2 TC03 CART-EC1 TC01 CART-EC2 TC01 --- # Minimum EP Test Suite for FR-07 TC Purpose ---- ----- TC01 Add new product to empty cart TC02 Add same product again TC03 Delete → Cancel TC04 Delete → Confirm TC05 Non-integer quantity TC06 Quantity < 1 This is the minimal equivalence-partitioning suite that satisfies the S04 selection guideline while covering every identified equivalence class at least once.</pre>		
Tool: ChatGPT Time: 20:36 07/06/2026 Prompt: " Now apply step 3 for FR-07 "	<pre>For FR-07, Step 3 = Boundary Value Analysis (BVA). According to S04, boundary testing should be applied to equivalence classes derived from ranges of values, testing values at and around the boundary (LB-1, LB, LB+1, etc.). --- # 1. Identify Range-Based Conditions Among the FR-07 ECs, only the Quantity field has a numeric range. From FR-06: > Quantity must be a positive integer, minimum value = 1. This produced: _ Valid EC: QTY-EC3 → Quantity ≥ 1 _ Invalid EC: QTY-EC4 → Quantity < 1 Boundary:</pre>	INCOMPLETE	The AI correctly follow guidelines(S04_Domai Testing.pdf) for BVA. However, it missed the test objective of the te cases.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning														
	_ LB = 1 No upper bound is specified. --- # 2. Existing EP Coverage From the EP suite: <table><tr><th>TC</th><th>Quantity</th></tr><tr><td>TC01</td><td>1</td></tr><tr><td>TC06</td><td>0</td></tr></table> Already covered: _ LB (1) _ LB-1 (0) However, S04 recommends testing immediately around the boundary. Therefore we should explicitly test: <table><tr><th>Position</th><th>Value</th></tr><tr><td>LB-1</td><td>0</td></tr><tr><td>LB</td><td>1</td></tr><tr><td>LB+1</td><td>2</td></tr></table> --- # 3. Additional Boundary Test Cases Assume Product A exists and all other inputs are valid. --- ## BVA-TC01 — Quantity = 0 (LB-1) ### Preconditions Product page is opened. ### Steps 1. Enter quantity = 0. 2. Click Add to Cart. ### Expected Result _ System rejects the quantity. _ Product is not added to cart. ### Boundary Covered _ LB-1 _ QTY-EC4 --- ## BVA-TC02 — Quantity = 1 (LB) ### Preconditions Product page is opened.	TC	Quantity	TC01	1	TC06	0	Position	Value	LB-1	0	LB	1	LB+1	2		
TC	Quantity																
TC01	1																
TC06	0																
Position	Value																
LB-1	0																
LB	1																
LB+1	2																

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning						
	### Steps 1. Enter quantity = 1. 2. Click Add to Cart. ### Expected Result _ Product added successfully. _ Quantity displayed as 1. ### Boundary Covered _ LB _ QTY-EC3 --- ## BVA-TC03 — Quantity = 2 (LB+1) ### Preconditions Product page is opened. ### Steps 1. Enter quantity = 2. 2. Click Add to Cart. ### Expected Result _ Product added successfully. _ Quantity displayed as 2. ### Boundary Covered _ LB+1 _ QTY-EC3 --- # 4. Additional UI Boundary for "-" Button FR-07 specifies quantity adjustment via + / - buttons. A common boundary risk is decrementing at the minimum quantity. --- ## BVA-TC04 — Click "-" When Quantity = 1 ### Preconditions Cart contains: <table><tr><td> Product </td><td> Quantity </td></tr><tr><td> ----- </td><td> ----- </td></tr><tr><td> Product A </td><td> 1 </td></tr></table> ### Steps 1. Click "-" once. ### Expected Result One of the following must happen: _ Quantity remains 1, or	Product	Quantity	-----	-----	Product A	1		
Product	Quantity								
-----	-----								
Product A	1								

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning
	<pre>_ System prevents decrementing below 1. But quantity must never become 0 or negative because the minimum allowed quantity is 1. ### Boundary Covered _ Transition attempt from LB to LB-1 --- # Final BVA Suite for FR-07 TC ID Quantity Boundary Position Expected ----- ----- ----- ----- BVA-TC01 0 LB-1 Reject BVA-TC02 1 LB Accept BVA-TC03 2 LB+1 Accept BVA-TC04 1 → click "-" Attempt to cross LB Prevent quantity < 1 These are the boundary-value tests that should be added on top of the equivalence-partition test suite for FR-07. They focus on the only numeric boundary present in the specification: the minimum quantity of 1.</pre>		
Tool: ChatGPT Time: 20:51 07/06/2026 Prompt: "Next, apply step 1 (requirements understanding and ECs identification) for FR-12"	<pre>Let's analyze FR-12: Access Control exactly as we did for FR-01 and FR-07. FR-12 states: > _ Admin module is only for accounts with role='admin'. > _ All Admin APIs (/api/admin/*) and data-modifying APIs must require: > > 1. A valid JWT Token. > 2. role='admin' in the Token. --- # Step 1 — Identify Inputs / Conditions The behavior of the API depends on two input conditions: ### C1. JWT Token validity The request may contain: _ A valid JWT token _ An invalid JWT token _ No token The system handles them differently. --- ### C2. Role contained in JWT If the token is valid, the role may be: _ admin _ non-admin (e.g., user) The system handles them differently. --- # Requirement R1: JWT Token must be valid This is a set-of-values condition. The specification explicitly distinguishes:</pre>	VALID	The AI correctly identifies the input fields and the requirements for FR-12, specifications based on the course guidelines(S04_Domain Testing.pdf). The equivalence classes provided by the AI were correct and suitable for test case design.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning
	<pre>_ valid token _ invalid token Also, from an API perspective, "missing token" is usually handled differently from "invalid token" (401 vs malformed/expired token), so according to the S04 guideline: > If elements are not handled identically, split into separate ECs. --- ## ECs for Token EC ID Equivalence Class Validity ----- ----- ----- TOK-EC1 Valid JWT token supplied Valid TOK-EC2 No JWT token supplied Invalid TOK-EC3 Invalid JWT token supplied (expired, malformed, tampered, etc.) Invalid --- # Requirement R2: Role must be admin This is a set-of-values condition. The specification explicitly distinguishes: _ role = admin _ role ≠ admin --- ## ECs for Role EC ID Equivalence Class Validity ----- ----- ----- ROLE-EC1 role = admin Valid ROLE-EC2 role ≠ admin Invalid Examples: _ user _ customer _ any non-admin role --- # Requirement R3: Admin API Access FR-12 applies to: 1. /api/admin/* 2. Data-modifying APIs: * POST /api/products * PUT /api/products * DELETE /api/products * /api/categories * /api/coupons For Domain Testing, the API endpoint itself is not really an input domain requiring partitioning because the requirement applies uniformly to all protected endpoints.</pre>		

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning																																																							
	Therefore we do not create ECs for individual URLs. --- # Combined Valid and Invalid Domains The authorization rule is: Access allowed iff: * Token is valid * AND role = admin Thus: ### Valid Domain <table><tr><td> EC Combination</td><td> </td></tr><tr><td> -----</td><td> </td></tr><tr><td> TOK-EC1 + ROLE-EC1</td><td> </td></tr></table> --- ### Invalid Domains <table><tr><td> EC Combination</td><td> </td></tr><tr><td> -----</td><td> </td></tr><tr><td> TOK-EC2 (missing token)</td><td> </td></tr><tr><td> TOK-EC3 (invalid token)</td><td> </td></tr><tr><td> TOK-EC1 + ROLE-EC2</td><td> </td></tr></table> Each should be tested separately because each represents a distinct authorization failure scenario. --- # Final EC List for FR-12 <table><tr><td> EC ID</td><td> Description</td><td> Validity</td></tr><tr><td> -----</td><td> -----</td><td> -----</td></tr><tr><td> TOK-EC1</td><td> Valid JWT token</td><td> Valid</td></tr><tr><td> TOK-EC2</td><td> Missing JWT token</td><td> Invalid</td></tr><tr><td> TOK-EC3</td><td> Invalid JWT token</td><td> Invalid</td></tr><tr><td> ROLE-EC1</td><td> role = admin</td><td> Valid</td></tr><tr><td> ROLE-EC2</td><td> role ≠ admin</td><td> Invalid</td></tr></table> --- # Minimal Domain Model The access-control decision can be represented as: <table><tr><td> Token</td><td> Role</td><td> Expected</td></tr><tr><td> -----</td><td> -----</td><td> -----</td></tr><tr><td> Valid</td><td> Admin</td><td> Allow</td></tr><tr><td> Missing</td><td> N/A</td><td> Deny</td></tr><tr><td> Invalid</td><td> N/A</td><td> Deny</td></tr><tr><td> Valid</td><td> Non-admin</td><td> Deny</td></tr></table> This gives us exactly the equivalence classes needed for Step 2 (test-case design) according to the S04 Domain Testing approach.	EC Combination		-----		TOK-EC1 + ROLE-EC1		EC Combination		-----		TOK-EC2 (missing token)		TOK-EC3 (invalid token)		TOK-EC1 + ROLE-EC2		EC ID	Description	Validity	-----	-----	-----	TOK-EC1	Valid JWT token	Valid	TOK-EC2	Missing JWT token	Invalid	TOK-EC3	Invalid JWT token	Invalid	ROLE-EC1	role = admin	Valid	ROLE-EC2	role ≠ admin	Invalid	Token	Role	Expected	-----	-----	-----	Valid	Admin	Allow	Missing	N/A	Deny	Invalid	N/A	Deny	Valid	Non-admin	Deny		
EC Combination																																																										

TOK-EC1 + ROLE-EC1																																																										
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TOK-EC2 (missing token)																																																										
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TOK-EC1 + ROLE-EC2																																																										
EC ID	Description	Validity																																																								
-----	-----	-----																																																								
TOK-EC1	Valid JWT token	Valid																																																								
TOK-EC2	Missing JWT token	Invalid																																																								
TOK-EC3	Invalid JWT token	Invalid																																																								
ROLE-EC1	role = admin	Valid																																																								
ROLE-EC2	role ≠ admin	Invalid																																																								
Token	Role	Expected																																																								
-----	-----	-----																																																								
Valid	Admin	Allow																																																								
Missing	N/A	Deny																																																								
Invalid	N/A	Deny																																																								
Valid	Non-admin	Deny																																																								

Tool: ChatGPT Time: 20:55 07/06/2026 Prompt: "Now, apply step 2 for FR-12"	Applying the S04 test case selection rule: > For valid classes, choose test cases to cover as many valid ECs as possible. > > For invalid classes, choose test cases so that each covers one and only one invalid class.	INCOMPLETE	The AI missed the test objective and test step for the test cases. The preconditions of test case TC03 should be te
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(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning																		
	The identified ECs are: <table><tr><td>EC</td><td>Description</td></tr><tr><td>TOK-EC1</td><td>Valid JWT token</td></tr><tr><td>TOK-EC2</td><td>Missing JWT token</td></tr><tr><td>TOK-EC3</td><td>Invalid JWT token</td></tr><tr><td>ROLE-EC1</td><td>role = admin</td></tr><tr><td>ROLE-EC2</td><td>role ≠ admin</td></tr></table> --- # Step 1: Valid Test Case The only valid authorization domain is: _ TOK-EC1 _ ROLE-EC1 --- ## TC01 – Admin Access with Valid Token ### Preconditions Admin account exists: _ admin@eshop.com _ role = admin Valid JWT token obtained after login. ### Request http
GET /api/admin/users
Authorization: Bearer <valid_admin_token>
 ### EC Covered <table><tr><td>EC</td><td></td></tr><tr><td>TOK-EC1</td><td></td></tr><tr><td>ROLE-EC1</td><td></td></tr></table> ### Expected Result _ Request succeeds. _ HTTP 200 OK. _ Admin resource returned. --- # Step 2: Invalid Test Cases Each test case should contain exactly one invalid EC. --- ## TC02 – Missing JWT Token ### Purpose Cover: _ TOK-EC2 only	EC	Description	TOK-EC1	Valid JWT token	TOK-EC2	Missing JWT token	TOK-EC3	Invalid JWT token	ROLE-EC1	role = admin	ROLE-EC2	role ≠ admin	EC		TOK-EC1		ROLE-EC1			input instead of test preconditions.
EC	Description																				
TOK-EC1	Valid JWT token																				
TOK-EC2	Missing JWT token																				
TOK-EC3	Invalid JWT token																				
ROLE-EC1	role = admin																				
ROLE-EC2	role ≠ admin																				
EC																					
TOK-EC1																					
ROLE-EC1																					

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning
	### Request http
GET /api/admin/users
 (No Authorization header) ### EC Covered EC ----- TOK-EC2 ### Expected Result _ Access denied. _ Authentication error returned. _ HTTP 401 Unauthorized (or equivalent). --- ## TC03 – Invalid JWT Token ### Purpose Cover: _ TOK-EC3 only ### Preconditions Use malformed, expired, or tampered token. ### Request http
GET /api/admin/users
Authorization: Bearer invalid_token
 ### EC Covered EC ----- TOK-EC3 ### Expected Result _ Access denied. _ Invalid token error. _ HTTP 401 Unauthorized. --- ## TC04 – Valid Token but Non-Admin Role ### Purpose Cover: _ ROLE-EC2 only ### Preconditions User account exists: _ test@eshop.com _ role = user Obtain a valid JWT token for this account.		

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning
	<pre>### Request http
GET /api/admin/users
Authorization: Bearer <valid_user_token>
 ### EC Covered EC ----- ROLE-EC2 ### Expected Result _ Access denied. _ Authorization error. * HTTP 403 Forbidden (or equivalent). --- # Coverage Matrix EC Covered By ----- ----- TOK-EC1 TC01 TOK-EC2 TC02 TOK-EC3 TC03 ROLE-EC1 TC01 ROLE-EC2 TC04 --- # Minimum EP Test Suite for FR-12 TC ID Scenario Expected Result ---- ----- TC01 Valid admin token Access granted TC02 Missing token Access denied TC03 Invalid token Access denied TC04 Valid token, non-admin role Access denied This is the minimum equivalence-partitioning test suite that covers every identified equivalence class exactly once while following the S04 selection guideline.</pre>		
Tool: ChatGPT Time: 20:58 07/06/2026 Prompt: "Now, apply step 3 for FR-12"	<pre>For Step 3 (Boundary Value Analysis), we should first check whether FR-12 contains any range-based input conditions, because BVA is designed for domains with boundaries (numeric ranges, lengths, dates, counts, etc.). --- # 1. Examine the FR-12 Conditions FR-12 contains two authorization conditions: 1. JWT Token must be valid. 2. Role must be <code>admin</code>. The identified ECs were: EC Description ----- ----- TOK-EC1 Valid token TOK-EC2 Missing token TOK-EC3 Invalid token ROLE-EC1 role = admin ROLE-EC2 role ≠ admin </pre>	VALID	The AI correctly analyz that there were no range-based inputs for FR-12, therefore Boundary Value Analys could not be applied.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning								
	--- # 2. Are There Any Boundaries? According to S04, BVA is applied to conditions such as: _ $1 \leq \text{count} \leq 999$ _ $3 \leq X \leq 7$ _ Length between 3 and 15 where explicit lower/upper limits exist. FR-12 does not specify: _ numeric ranges, _ string length limits, _ date limits, _ quantity limits, _ count limits. Instead, it uses discrete sets of values: _ Valid vs Invalid token _ Admin vs Non-admin role These are equivalence partitions, not ranges. --- # 3. Result ### Strict Domain Testing Answer No additional Boundary Value Analysis test cases are applicable to FR-12. Reason: _ There is no input condition with a lower or upper boundary. _ All identified conditions are set-of-values conditions. _ The EP suite already completely covers the domain. Therefore: <table><tr><td> Step</td><td> Applicable? </td></tr><tr><td> -----</td><td> ----- </td></tr><tr><td> Equivalence Partitioning Yes</td><td> </td></tr><tr><td> Boundary Value Analysis No</td><td> </td></tr></table> --- # 4. Common Mistake Students sometimes try to invent boundaries such as: _ JWT length _ Token expiration time _ Role string length However, these boundaries are not stated in FR-12 and therefore should not be introduced during specification-based Domain Testing. According to the S04 approach, boundaries must come from the specified input conditions, not assumptions about implementation details. ### Final Conclusion For FR-12:	Step	Applicable?	-----	-----	Equivalence Partitioning Yes		Boundary Value Analysis No			
Step	Applicable?										
-----	-----										
Equivalence Partitioning Yes											
Boundary Value Analysis No											

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning
	_ Step 1 (EC Identification): applicable ✓ _ Step 2 (EP Test Design): applicable ✓ * Step 3 (BVA): not applicable because FR-12 contains no range-based input domain and therefore has no meaningful boundaries to test.		
Tool: ChatGPT Time: 20:49 09/06/2026 Prompt: "Create an agent skill to use in vscode with Copilot to perform Domain testing. The skill first takes a README contains functional specifications as the input. Then the skill asks the user for the feature, or the functional requirement that the user wants to perform domain testing on and execute step 1 (understand requirements and identify ECs), step 2 (design test cases) and step 3(design BVA test cases, if possible, if not explain clearly why). The skill already contains the guidelines from the slides for all the steps. The requirement understanding step is THE MOST IMPORTANT step, make sure that the skill identifies the requirements of the inputs correctly and fully, and MUST NOT make any assumptions on behalf of the specifications. You MUST go to https://code.visualstudio.com/docs/agent-customization/agent-skills, find the 'To create a skill:' section to know how to format the file. Give the direct file content here, presented as a code block. You should include the result of step 1, step 2 and step 3 of FR-01 as examples."	Below is a complete SKILL.md content that follows the VS Code Agent Skills format (YAML frontmatter + instructions body). Skills are defined by a folder containing a SKILL.md file. ([Visual Studio Code][1]) <pre> markdown

name: domain-testing
description: Use when performing Equivalence Partitioning (EP) and Boundary Value Analysis (BVA) from functional requirements in a README, SRS, PRD, user stories, or software specification. Analyzes a selected functional requirement, identifies equivalence classes, designs minimal EP test cases, and designs BVA test cases when applicable.

Domain Testing Skill

Purpose

This skill performs Domain Testing according to the standard workflow:

1. Requirement Understanding
2. Equivalence Class (EC) Identification
3. Equivalence Partitioning (EP) Test Design
4. Boundary Value Analysis (BVA)

This skill is intended for software testing, QA, test design, and test case generation from specifications.

REQUIRED WORKFLOW

When invoked:

Step 0 - Obtain Inputs

First identify:

1. The specification document (README, SRS, PRD, etc.)
2. The Functional Requirement (FR) to analyze

If the requirement is not specified by the user, ask:

> Which Functional Requirement (FR) do you want me to perform Domain Testing on?

Do not proceed until a specific FR is selected.

CRITICAL RULES

Rule 1 - Specification First

ONLY use information explicitly stated in the specification.

Never invent:

- constraints
- limits
- ranges
- validations
- formats
- business rules

that are not explicitly documented.

If something is not stated:

State clearly:

> This requirement is not specified. No equivalence classes can be derived from it.

Never assume implementation details.

Examples of forbidden assumptions:

- maximum password length
- JWT expiration duration
- HTTP status codes
- database schema constraints
- UI behavior not stated in the specification

Rule 2 - Requirement Understanding is the Most Important Step

Before identifying ECs:

Extract ALL requirements affecting input behavior.

For every input field:

1. Identify the field.
2. Identify every condition applied to that field.
3. Classify each condition.

Do NOT jump directly into EC identification.

Always show:

Input Field

Requirement

Condition Type

Where Condition Type is one of:

- Range of values
- Set of values
- Must be condition

DOMAIN TESTING GUIDELINES

Use the following rules.

Range of Values

Examples:

- 1 <= count <= 999
- length between 3 and 15
- quantity >= 1

Create:

- 1 valid EC
- 2 invalid ECs

Pattern:

Valid:
- within range

Invalid:
- below range
- above range

If only one boundary exists:

Example:

quantity >= 1

Create:

Valid:
- quantity >= 1

Invalid:
- quantity < 1

Set of Values

Example:

Vehicle type must be:

- BUS
- TRUCK
- TAXI
- MOTORCYCLE

If values are handled differently:

Create one valid EC per meaningful value/group.

Also create invalid ECs when appropriate.

Split ECs further whenever behavior differs.

Must Be Condition

Examples:

- Must be an email
- Must be an integer
- First character must be a letter
- Password must contain uppercase

Create:

Valid:
- condition satisfied

Invalid:
- condition violated

EC Splitting Rule
 </pre>	VALID	The skill correctly summarized the guidelines within the slides(S04_Domain Testing.pdf) and gave clear instruct on how to perform each step. In addition, the skill also included examples from the previous working domain testing on the first feature.

(1) Prompt + Tool

(2) AI Output

(3) Verdict

(4) Reasoning

If there is reason to believe values are handled differently by the system:

Split into separate ECs.

Example:

Email uniqueness:

Valid:
- email does not exist

Invalid:
- email already exists

Even though both are valid email formats.

STEP 1 - REQUIREMENT UNDERSTANDING & EC IDENTIFICATION

Produce:

Requirement Analysis

For each field:

Field

Requirement

Condition Type

Equivalence Classes

Use format:

| EC ID | Description | Validity |
-----	-----	-----	 Afterward produce: ## Consolidated Equivalence Classes containing all ECs. --- # STEP 2 - TEST CASE DESIGN (EP) Use the following strategy. ## Valid Classes Create as few test cases as possible. Each valid test case should cover as many valid ECs as possible. Continue until every valid EC is covered. ## Invalid Classes Create one test case per invalid EC. Each invalid test case must contain: - exactly one invalid EC - all other inputs valid Do not combine multiple invalid ECs in the same test. --- ## Output Format For every test case: 	TC ID	Inputs	EC Covered	Expected Result	 Then provide: ## Coverage Matrix 	EC	Covered By	 Verify every EC is covered. --- # STEP 3 - BOUNDARY VALUE ANALYSIS Apply BVA ONLY when a true boundary exists. Examples: - length >= 8 - 1 <= quantity <= 999 - 3 <= x <= 7 For each boundary: Generate: - LB-1 - LB - LB+1 and if an upper boundary exists: - UB-1 - UB - UB+1 Use the standard BVA approach. --- ## When BVA Does NOT Apply If requirements only contain: - valid/invalid token - admin/non-admin role - existing/non-existing item - yes/no decisions then state: > Boundary Value Analysis is not applicable because the specification contains no range-based input domain and therefore no meaningful boundaries. Do NOT invent boundaries. --- # OUTPUT STRUCTURE Always produce: # Step 1 - Requirement Understanding and EC Identification ... # Step 2 - Equivalence Partitioning Test Cases ... # Step 3 - Boundary Value Analysis ... # Final Coverage Summary ... --- # EXAMPLE The following example demonstrates the expected level of detail. ===== EXAMPLE: FR-01 Registration ===== Requirement: - Full Name required - Email required - Email format valid - Email unique - Password required - Password length >= 8 - Password contains uppercase - Password contains lowercase - Password contains digit - Password contains special character - Confirm Password required - Confirm Password matches Password ----- STEP 1 ----- ----- ## ECs ### Full Name 	EC ID	Description	Validity
-----	-----	-----	 	FN-EC1	Full Name provided	Valid								
FN-EC2	Full Name empty	Invalid	 ### Email Required 	EC ID	Description	Validity								
-----	-----	-----	 	EM-EC1	Email provided	Valid								
EM-EC2	Email empty	Invalid	 ### Email Format 	EC ID	Description	Validity								
-----	-----	-----	 	EM-EC3	Valid email format	Valid								
EM-EC4	Invalid email format	Invalid	 ### Email Uniqueness 	EC ID	Description	Validity								
-----	-----	-----	 	EM-EC5	Email does not exist	Valid								
EM-EC6	Email already exists	Invalid	 ### Password Required 	EC ID	Description	Validity								
-----	-----	-----	 	PW-EC1	Password provided	Valid								
PW-EC2	Password													

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning
	<pre>empty Invalid

Password Length

 EC ID Description Validity
 ----- ----- -----
 PW-EC3 Length >= 8 Valid
 PW-EC4 Length < 8 Invalid

Password Uppercase

 EC ID Description Validity
 ----- ----- -----
 PW-EC5 Contains uppercase Valid
 PW-EC6 No uppercase Invalid

Password Lowercase

 EC ID Description Validity
 ----- ----- -----
 PW-EC7 Contains lowercase Valid
 PW-EC8 No lowercase Invalid

Password Digit

 EC ID Description Validity
 ----- ----- -----
 PW-EC9 Contains digit Valid
 PW-EC10 No digit Invalid

Password Special Character

 EC ID Description Validity
 ----- ----- -----
 PW-EC11 Contains allowed special character Valid
 PW-EC12 No special character Invalid
 PW-EC13 Unsupported special character only Invalid

Confirm Password Required

 EC ID Description Validity
 ----- ----- -----
 CP-EC1 Confirm Password provided Valid
 CP-EC2 Confirm Password empty Invalid

Confirm Password Match

 EC ID Description Validity
 ----- ----- -----
 CP-EC3 Matches Password Valid
 CP-EC4 Does not match Password Invalid

----- -
STEP 2

Valid test:

TC01

Full Name:
Nguyen Van A

Email:
newuser@gmail.com

Password:
Abc1234@

Confirm:
Abc1234@

Covers:

FN- EC1
EM-EC1
EM-EC3
EM-EC5
PW-EC1
PW-EC3
PW- EC5
PW-EC7
PW-EC9
PW-EC11
CP-EC1
CP-EC3

Expected:
Registration successful.

Invalid tests:

TC02:
FN-EC2 only

TC03:
EM-EC2 only

TC04:
EM-EC4 only

TC05:
EM-EC6 only

TC06:
PW-EC2 only

TC07:
PW-EC4 only

TC08:
PW-EC6 only

TC09:
PW-EC8 only

TC10:
PW-EC10 only

TC11:
PW-EC12 only

TC12:
PW-EC13 only

TC13:
CP-EC2 only

TC14:
CP-EC4 only

----- -----
STEP 3
----- -----

Boundary:
Password length >= 8

LB = 8

BVA-TC01:
Length = 7
Expected: Reject

BVA-TC02:
Length = 8
Expected: Accept

BVA-TC03:
Length = 9
Expected: Accept

FINAL CONCLUSION
----- -----

- All ECs identified
- All ECs covered by EP tests
- BVA applied only to documented boundaries
- No assumptions introduced beyond the specification
</pre> Suggested location: <pre>text
.agents/skills/domain-testing/SKILL.md
</pre> This follows the VS Code Agent Skills structure: a skill directory containing a SKILL.md file with YAML frontmatter (name, description) and instructions that define the workflow. ([Visual Studio Code][1]) [1]: https://code.visualstudio.com/docs/copilot/customization/agent-skills?pubDate=20260108&utm_source=chatgpt.com "Use Agent Skills in VS Code"		

3. Summary of AI Accuracy

Metric	Count	Percentage
Total AI-generated artifacts audited	10	
VALID (correct, accepted as-is)	4	40%

Metric	Count	Percentage
INVALID (wrong; rejected)	0	0%
INCOMPLETE (acceptable after edits)	6	60%

4. Conclusion

In conclusion, the AI tool provided great support for applying domain testing and boundary value analysis in designing test cases for the features of EShop. While the tool correctly identified most of the equivalence classes, there were instances where the AI noticed some of the requirements were not suitable to be used for domain testing, yet still chose to create equivalence classes based on such requirements. In addition to that, the AI tool missed test objective for all of the test case design and missed test step for some test cases. For future usages, my recommendation is trying to understand what kind of outcome is wanted and predict beforehand how the AI might produce that outcome.

5. Mandatory Disclosure

Equivalence classes, test case design and BVA test cases was initially generated by ChatGPT; I reviewed and modified test case design for FR-01, FR-07 and FR-12, added test objectives and test steps; the bug report and test summary report was written entirely by me. The detailed AI Audit Report is attached as Appendix A. I confirm I did not use AI to generate any artifact listed in the prohibited category.

Signature

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- **Date:** 10/06/2026
- **Signature:**



Reference

- Kharbach, M. (2026). AI Use Policy Templates for Higher Education. CC BY-NC-SA 4.0.
- ISTQB Foundation Level Syllabus (latest version).
- Hardman, P. (2025). A Post-AI Learning Taxonomy.
- Fuster Rabella, M. (2025). OECD Education Working Paper No. 338.
- Perkins, M., Roe, J., & Furze, L. (2025). AI Assessment Scale.
- Anthropic (2025). Building reliable AI test agents — engineering blog.
- DeepEval & Promptfoo documentation — testing frameworks for LLM systems.